

ALTNAHARRA, United Kingdom

WMO: 030440

Lat: 58.2878N

Lon: 4.4425W

Elev: 80

StdP: 100.37

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-9.9	-6.7	-10.5	1.6	-8.0	-6.7	2.2	-4.6	16.9	6.0	14.8	6.4	0.4	320	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	22.7	16.5	20.6	15.4	18.9	14.5	17.5	21.2	16.3	19.5	15.2	18.1	4.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.0	11.5	19.1	14.7	10.6	18.0	13.8	9.9	16.8	49.4	21.4	46.0	19.4	42.9	18.2	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.3	9.7	-13.1	25.5	3.7	1.9	-15.8	26.9	-18.0	28.0	-20.1	29.1	-22.8	30.4		
(5)			-12.3	19.7	3.6	2.3	-14.9	21.4	-17.0	22.7	-19.1	24.0	-21.7	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	3.0	3.3	4.5	6.5	8.6	11.3	13.5	13.1	11.4	8.5	5.4	2.8	
	DBStd	5.00	3.76	3.57	3.28	3.17	3.12	2.75	2.47	2.33	2.62	3.08	3.54	4.73	
	HDD10.0	1219	217	189	171	111	66	17	2	3	16	66	141	223	
	HDD18.3	3889	474	422	429	354	303	210	151	163	209	306	388	480	
	CDD10.0	372	0	1	1	7	21	57	110	98	56	18	3	1	
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0	
	CDH23.3	38	0	0	0	0	4	7	19	8	1	0	0	0	
	CDH26.7	1	0	0	0	0	0	0	1	1	0	0	0	0	
Wind	WSAvg	3.9	4.4	4.4	4.5	4.1	3.8	3.7	3.2	3.3	3.6	4.0	4.0	3.8	
Precipitation	PrecAvg	1133	129	102	91	71	70	63	64	84	94	121	131	123	
	PrecMax	1363	235	181	171	126	134	108	111	172	263	266	195	224	
	PrecMin	899	18	30	27	26	3	28	24	38	31	34	64	21	
	PrecStd	127	69	43	40	22	34	21	27	32	48	57	39	52	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	12.7	14.7	19.3	23.4	24.1	25.8	24.6	21.9	17.2	14.5	13.2	
		MCWB	9.6	10.2	9.7	12.2	15.2	16.5	18.3	18.3	16.0	13.3	11.6	10.7	
	2%	DB	10.6	11.3	12.1	16.1	19.9	20.7	22.7	21.1	19.1	15.6	12.7	12.0	
		MCWB	8.6	9.4	8.9	11.1	13.4	14.8	17.4	16.5	14.7	12.6	10.5	10.0	
	5%	DB	9.6	10.1	10.7	14.0	17.2	18.3	20.4	19.1	17.4	14.4	11.6	10.9	
		MCWB	7.8	8.3	8.4	10.1	12.2	13.7	16.0	15.0	13.8	11.7	9.6	9.0	
	10%	DB	8.3	8.6	9.5	12.2	14.9	16.6	18.5	17.6	16.0	13.2	10.6	9.4	
		MCWB	6.7	6.7	7.4	9.1	11.0	12.8	15.0	14.2	12.9	10.8	8.8	7.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	10.8	10.8	13.5	16.8	17.4	19.6	19.3	16.9	14.3	12.2	11.3
			MCDB	11.6	12.1	12.8	17.6	21.0	22.8	23.4	23.0	20.0	16.1	13.9	12.7
2%		WB	9.0	9.7	9.6	11.6	14.4	15.5	17.7	17.0	15.4	13.1	10.9	10.3	
		MCDB	10.4	11.1	11.5	15.0	18.8	19.6	21.7	20.2	18.2	15.1	12.5	11.8	
5%		WB	7.9	8.5	8.6	10.6	12.6	14.3	16.6	15.7	14.4	12.1	9.8	9.2	
		MCDB	9.4	10.1	10.5	13.5	16.6	17.6	19.8	18.3	16.8	13.9	11.3	10.9	
10%		WB	6.8	6.9	7.6	9.4	11.3	13.2	15.4	14.6	13.4	11.2	8.9	7.8	
		MCDB	8.3	8.4	9.5	12.1	14.6	16.1	18.2	17.1	15.6	12.9	10.5	9.5	
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.6	7.3	8.7	9.7	8.2	8.0	7.9	8.1	7.0	6.1	6.4	
		MCDBR	6.1	7.2	9.2	12.5	14.9	12.2	12.4	11.0	11.2	8.2	6.6	7.1	
		MCWBR	5.4	5.7	6.3	7.6	9.3	7.2	7.6	6.8	7.2	5.9	5.5	6.1	
	5% WB	MCDBR	5.9	6.2	7.9	10.6	13.3	10.9	11.4	9.9	9.5	7.1	6.3	6.8	
		MCWBR	5.4	5.4	6.2	7.1	8.9	6.9	7.3	6.6	6.7	5.8	5.7	6.1	
Clear-Sky Solar Irradiance	taub	0.322	0.343	0.361	0.379	0.377	0.371	0.376	0.365	0.354	0.340	0.321	0.308		
	taud	2.162	2.261	2.294	2.291	2.350	2.403	2.403	2.441	2.459	2.387	2.216	2.107		
	Ebn at Noon	483	673	776	826	855	866	851	837	787	677	484	367		
	Edh at Noon	52	74	94	111	112	108	105	95	80	65	48	40		
All-Sky Solar Radiation	RadAvg	0.34	0.93	1.98	3.45	4.65	4.57	4.26	3.45	2.32	1.19	0.46	0.22		
	RadStd	0.04	0.12	0.15	0.23	0.32	0.41	0.32	0.20	0.16	0.11	0.05	0.02		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page